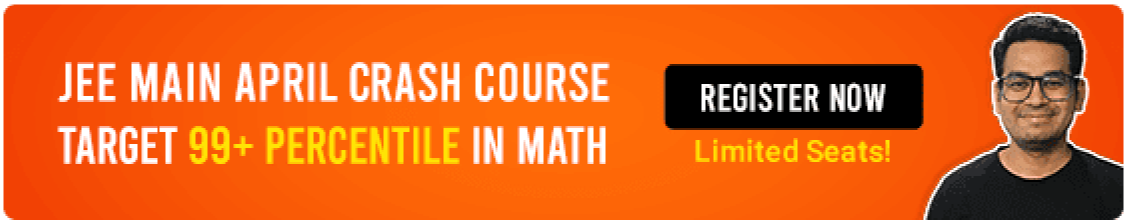
Master Math for JEE Main & JEE Advanced

Crash Courses designed specifically for students who want to improve their percentile & score in upcoming JEE Main & JEE Advanced exam. **Tap on the banners to know more.**

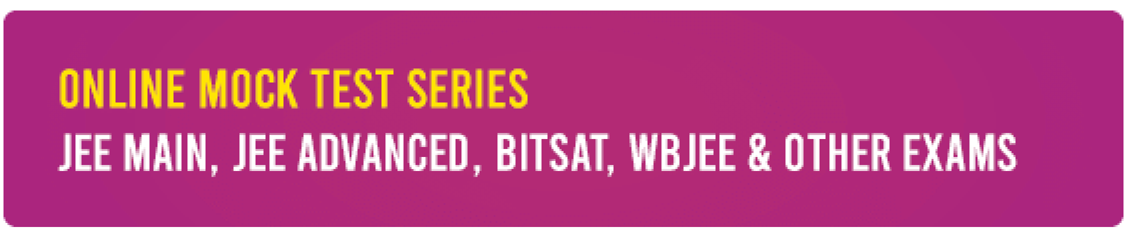
For JEE Main 2020 April



For JEE Advanced 2020

Online Mock Test Series

Take online mock tests for a solid practice for your upcoming JEE Main, JEE Advanced, BITSAT & other competitive exams. All the subjects are included. **Tap on the banner.**



Free Study Materials

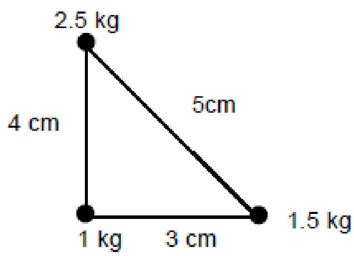
Download PDF of study materials like notes, formula sheets, practice questions, previous year question papers from our website. Visit www.mathongo.com now!

JEE Mains 2020 Jan Chapter wise Question Bank

Center of Mass, Momentum Conservation and Collision

Q1

Three point masses 1kg, 1.5 kg, 2.5 kg are placed at the vertices of a triangle with sides 3cm, 4cm and 5cm as shown in the figure. The location of centre of mass with respect to 1kg mass is :

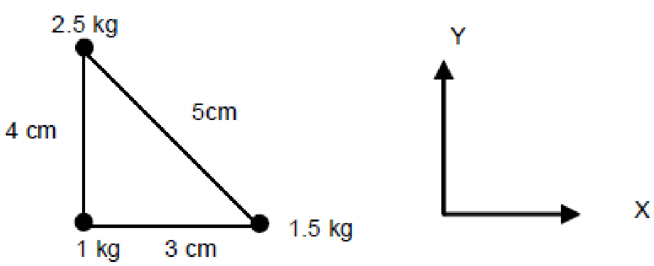


(1) 0.6 cm to the right of 1 kg and 2 cm above 1 kg mass
(2) 0.9 cm to the right of 1kg and 2 cm above 1 kg mass
(3) 0.9 cm to the left of 1kg and 2 cm above 1kg mass
(4) 0.9 cm to the right of 1 kg and 1.5 cm above 1kg mass

7th Jan Morning

Sol

(2)
Take 1kg mass at origin
1 kg द्रव्यमान को मूल बिन्दु लेने पर



$$X_{cm} = \frac{1\times 0 + 1.5\times 3 + 2.5\times 0}{5} = 0.9\text{cm}$$

$$Y_{cm} = \frac{1\times 0 + 1.5\times 0 + 2.5\times 4}{5} = 2\text{cm}$$

Q2

A solid cube of side 'a' is shown in the figure. Find maximum value of $100\frac{b}{a}$ for which the block does not topple before sliding.

7th Jan Evening

Sol

50.00

For no toppling नहीं पलटने के लिए

$$F\left(\frac{a}{2}+b\right) \le mg\frac{a}{2}$$

$$\mu\frac{a}{2}+\mu b \le \frac{a}{2}$$

$0.2a + 0.4b \le 0.5a$

$0.4b \le 0.3a$

$$b \le \frac{3a}{4}$$

$b \le 0.75a$ (in limiting case सीमान्त स्थिति में)

But it is not possible as b can maximum be equal to 0.5a

परन्तु यह सम्भव नहीं क्योंकि b का अधिकतम 0.5a हो सकता है।

$$\therefore \quad \left(100\frac{b}{a}\right)_{max.} = 50.00$$

Q3

A block of mass m is connected at one end of spring fixed at other end having natural length ℓ_0 and spring constant K. The block is rotated with constant angular speed (ω) about the fixed end in gravity free space. The elongation in spring is–

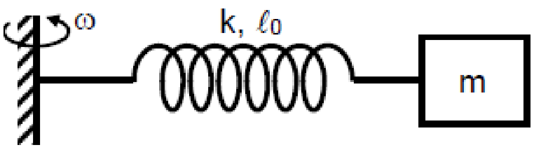
एक ब्लॉक जिसका द्रव्यमान m है को प्राकतिक लम्बाई ℓ_0 तथा बल नियतांक K की स्प्रिंग के एक सिरे से जुड़ा हुआ है तथा स्प्रिंग का दूसरा सिरा जड़वत् है। ब्लॉक को जड़वत सिरे के परितः गुरुत्वहीन स्थान में नियत कोणिय चाल (ω) से घुमाया जाता है तो स्प्रिंग मे खिचाव बताइए

(1) $\frac{\ell_0 m\omega^2}{k-m\omega^2}$ (2) $\frac{\ell_0 m\omega^2}{k+m\omega^2}$ (3) $\frac{\ell_0 m\omega^2}{k-m\omega}$ (4) $\frac{\ell_0 m\omega^2}{k+m\omega}$

8th Jan Morning

Sol

(1)



$m\omega^2(\ell_0 + x) = kx$

$$\left(\frac{\ell_0}{x}+1\right)=\frac{k}{m\omega^2}$$

$$x = \frac{\ell_0 m\omega^2}{k - m\omega^2}$$

Q4

A rod of mass 4m and length L is hinged at the mid point. A ball of mass 'm' moving with speed V in the plane of rod which is perpendicular to axis of rotation, strikes at one end at an angle of 45° and sticks to it. The angular velocity of system after collision is–

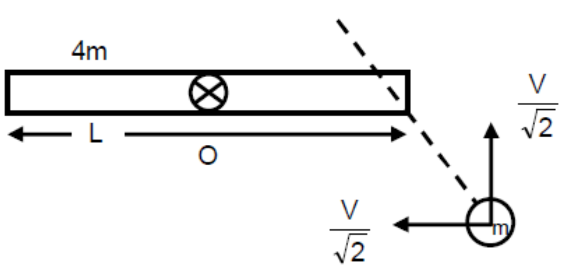
4m द्रव्यमान तथा L लम्बाई की छड़ मध्य बिन्दु से निलम्बित है। 'm' द्रव्यमान की गेंद छड़ के तल (जो कि घूर्णन अक्ष के लम्बवत् है) में V चाल से गति करती हुई छड़ के एक सिरे से 45° कोण पर टक्करा कर चिपक जाती है। टक्कर के पश्चात निकाय का कोणीय वेग ज्ञात कीजिए

(1) $\frac{3\sqrt{2}V}{7L}$ (2) $\frac{\sqrt{2}V}{7L}$ (3) $\frac{\sqrt{2}V}{3L}$ (4) $\frac{3V}{7L}$

8th Jan Morning

Sol

(1)



$L_{oi} = L_{of}$

$$\frac{mV}{\sqrt{2}}\times\frac{L}{2}=\left[\frac{4mL^2}{12}+\frac{mL^2}{4}\right]\times\omega$$

$$\omega=\frac{6V}{7\sqrt{2}L} = \frac{3\sqrt{2}V}{7L}$$

Q5

Two particles each of mass 0.10kg are moving with velocities 3m/s along x axis and 5m/s along y-axis respectively. After an elastic collision one of the mass moves with a velocity $4\hat{i}+4\hat{j}$. The energy of other particle after collision is $\frac{x}{10}$, then x is.

8th Jan Morning

Sol

1

For elastic collision प्रत्यास्थ टक्कर के लिये $KE_i = KE_f$

$$\frac{1}{2}m\times 25+\frac{1}{2}\times m\times 9=\frac{1}{2}m\times 32+\frac{1}{2}mv^2$$

$34 = 32 + v^2$

$KE = \frac{1}{2}\times 0.1\times 2 = 0.1J = \frac{1}{10}$

$x = 1$

Q6

There are two identical particles A and B. One is projected vertically upward with speed $\sqrt{2gh}$ from ground and other is dropped from height h along the same vertical line. Collision between them is perfectly inelastic. Find time taken by them to reach the ground after collision in terms of $\sqrt{\frac{h}{g}}$.

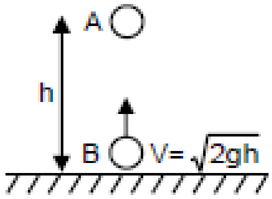
दो कण A तथा B दिये गये है। एक को $\sqrt{2gh}$ चाल से सतह से ऊपर की ओर फेंका जाता है तथा दूसरे को ऊँचाई h से समान ऊर्ध्वाधर रेखा के अनुदिश छोड़ा जाता है। यदि इनके मध्य टक्कर पूर्णतया अप्रत्यास्थ है तो टक्कर के पश्चात् निकाय को धरातल पर पहुँचने में लगा समय $\sqrt{\frac{h}{g}}$ के पदो में ज्ञात करो।

(1) $\sqrt{\frac{3}{2}}$ (2) $\sqrt{\frac{1}{2}}$ (3) $\sqrt{3}$ (4) $\sqrt{\frac{1}{5}}$

8th Jan Evening

Sol

(1)



time for collision टक्कर के लिए समय $t_1 = \frac{h}{\sqrt{2gh}}$

After t_1 के बाद

$V_A = 0 - gt_1 = -\sqrt{\frac{gh}{2}}$

and तथा $V_B = \sqrt{2gh} - gt_1 = \sqrt{gh}\left[\sqrt{2} - \frac{1}{\sqrt{2}}\right]$

at the time of collision
टक्कर के दौरान

$\vec{P}_i = \vec{P}_f$

$\Rightarrow m\vec{V}_A + m\vec{V}_B = 2m\vec{V}_f$

$\Rightarrow -\sqrt{\frac{gh}{2}} + \sqrt{gh}\left[\sqrt{2} - \frac{1}{\sqrt{2}}\right] = 2\vec{V}_f$

$V_f = 0$

and height from ground और धरातल से ऊँचाई $= h - \frac{1}{2}gt_1^2 = h - \frac{h}{4} = \frac{3h}{4}$

so time अतः समय = $\sqrt{2 \times \frac{\left(\frac{3h}{4}\right)}{g}} = \sqrt{\frac{3h}{2g}}$

Q7

Two particles of same mass 'm' moving with velocities $\vec{v}_1 = v\hat{i}$ and $\vec{v}_2 = \frac{v}{2}\hat{i} + \frac{v}{2}\hat{j}$ collide in-elastically. Find the loss in kinetic energy.

समान द्रव्यमान 'm' के दो गतिशील कणो के वेग क्रमशः $\vec{v}_1 = v\hat{i}$ तथा $\vec{v}_2 = \frac{v}{2}\hat{i} + \frac{v}{2}\hat{j}$ है। दोनों अप्रत्यास्थ रूप से टकराते है। गतिज ऊर्जा में हानि ज्ञात करो–

(1) $\frac{mv^2}{8}$ (2) $\frac{5mv^2}{8}$ (3) $\frac{mv^2}{4}$ (4) $\frac{3mv^2}{8}$

9th Jan Morning

Sol

(1)

Conserving momentum संवेग सरंक्षण से

$$mv\hat{i} + m\left(\frac{v}{2}\hat{i} + \frac{v}{2}\hat{j}\right) = 2m(v_1\hat{i} + v_2\hat{j})$$

on solving हल करने पर

$v_1 = \frac{3v}{4}$ and $v_2 = \frac{v}{4}$

Change in K.E. गतिज ऊर्जा में परिवर्तन

$$\left[\frac{1}{2}mv^2 + \frac{1}{2}m\left(\frac{v}{2}\sqrt{2}\right)^2\right] - \left[\frac{1}{2}(2m)\left(\frac{9v^2}{16} + \frac{v^2}{16}\right)\right]$$

$$= \frac{3mv^2}{4} - \frac{5mv^2}{8} = \frac{mv^2}{8}$$

Q8

A particle is projected from the ground with speed u at angle 60° from horizontal. It collides with a second particle of same mass moving with horizontal speed u in same direction at highest point of its trajectory. If collision is perfectly inelastic then find horizontal distance travelled by them after collision when they reached at ground

एक कण को धरातल से u चाल से 60° के कोण पर प्रेक्षेपित किया जाता है। यह अपने उच्चतम बिन्दु पर क्षैतिज दिशा में u चाल से समान दिशा में गतिशील समान द्रव्यमान के कण से पूर्णतः अप्रत्यास्थ टक्कर करता है, तो इनके द्वारा टक्कर के पश्चात् धरातल तक पहुचनें में चली गयी क्षैतिज दूरी ज्ञात करो–

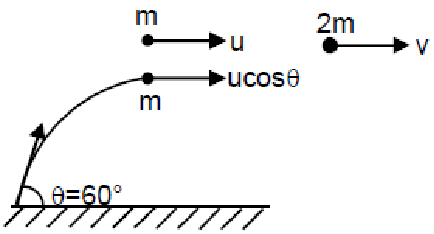
(1) $\frac{3\sqrt{6}u^2}{8g}$ (2) $\frac{3\sqrt{3}u^2}{8g}$ (3) $\frac{u^2}{8g}$ (4) $\frac{\sqrt{3}u^2}{g}$

(3)

9th Jan Evening

Sol

(2)



$p_i = p_f$

$mu + mu\cos\theta = 2mv$

$\Rightarrow v = \frac{u(1+\cos 60°)}{2} = \frac{3}{4}u$

so horizontal range after collision टक्कर के पश्चात् क्षैतिज परास = vt

$$= v\sqrt{\frac{2H_{max}}{g}}$$

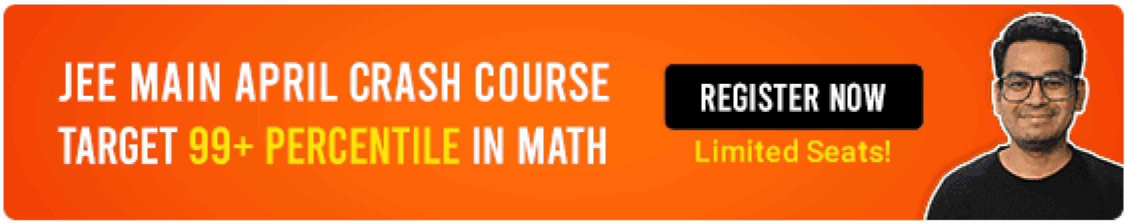
$$= \frac{3}{4}u\sqrt{\frac{2u^2\sin^2(60°)}{2g^2}}$$

$$= \frac{3}{4}u^2\frac{\sqrt{\frac{3}{4}}}{g} = \frac{3\sqrt{3}u^2}{8g}$$

Master Math for JEE Main & JEE Advanced

Crash Courses designed specifically for students who want to improve their percentile & score in upcoming JEE Main & JEE Advanced exam. **Tap on the banners to know more.**

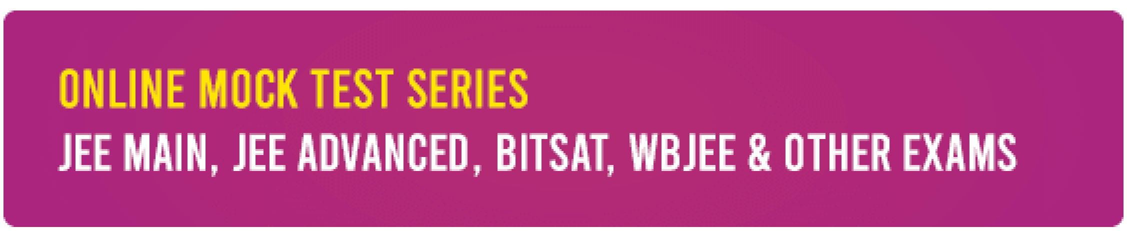
For JEE Main 2020 April



For JEE Advanced 2020

Online Mock Test Series

Take online mock tests for a solid practice for your upcoming JEE Main, JEE Advanced, BITSAT & other competitive exams. All the subjects are included. **Tap on the banner.**



Free Study Materials

Download PDF of study materials like notes, formula sheets, practice questions, previous year question papers from our website. Visit www.mathongo.com now!